

Pundarikaksh N. Tripathi

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Experience

FlyRank AI | Causal ML, X-Learner, Random Forest, scikit-learn, DuckDB, Hugging Face Datasets, Python

Remote

Machine Learning Engineering Intern

July 2026 – September 2026

- Built a refresh-opportunity ranker on 30,000 anonymized pages: dropped label-derived columns (`trend_pct`, `trend_direction`), trained a Random Forest on a client-grouped 70/30 split, raising Precision@50 from 0.56 (hand-written rule) to 0.70.
- Quantified leakage: random-split P@50 0.94 vs 0.70 under client-grouped holdout (0.24 optimism gap); reframed claims as observational.
- Capstone (mentor-accepted): X-Learner estimating the CATE of a content refresh (random-forest learners, decile Qini on a client-grouped holdout); ran DuckDB `hf://` Parquet queries over the full 79.8M-row warehouse.
- Built an action playbook (reason codes, no-go list); documented the unobserved-confounding limit and the need for an A/B test.

Projects

🔗 **Aegis: Behavioral Governance for Autonomous Agents (OpenAI Buildweek)** | Go, C, eBPF LSM, seccomp, AWS Cedar, SQLite, LinUCB, RL

- Designed and built (one-week sprint; teammates helped with presentation and live testing) a Linux control plane for AI coding agents: C eBPF LSM hooks on `file_open`, `socket_connect` and `bprm_check_security` feed a ring buffer read by a Go daemon.
- Scored event sequences with a temporal graph (zero-trust invariants plus an EMA baseline); flagged events go to a retrieval-augmented adjudicator: exact SHA-256 recall for `exec`, cosine > 0.95 recall otherwise, LLM cascade only on a miss.
- Compiled stored denials (a Cedar `forbid` subset) into BPF map entries for in-kernel blocking; added a seccomp user-notification `exec` gate, a Bubble Tea TUI and an offline LinUCB bandit (FN –5, FP –1).
- Scoring costs 3.8 μ s/event; float32-BLOB embeddings cut episodic recall from 278 ms to 11 ms at 5,000 cases; 57 Go tests under the race detector; CI compiles the eBPF objects and gates the evals.
- Validated live on Linux and WSL2 in a recorded demo: blocked the post-exploitation payload of BunnyHijack (a disputed Bun package-manager PoC) at the syscall layer, plus other unauthorised commands.

🔗 **Source:** github.com/quiet-intelligence/aegis

🔗 **Causal-DML: Counterfactual Churn Engine on KKBox Logs** | Python, DuckDB, EconML, DoWhy, FastAPI, MLflow, Double ML, Streamlit

- Aggregated the 30.5 GB KKBox `user_logs.csv` into per-user features with one in-process DuckDB query (`read_csv_auto`, GROUP BY), no cluster.
- Estimated heterogeneous effects with DoWhy and EconML LinearDML: random-forest nuisances, 3-fold cross-fitting, Neyman-orthogonal residual regression on 100k users, tracked in MLflow.
- Validated with a placebo design (synthetic confounded treatment on real churn labels, true effect zero): ATE –0.0017, outcome R^2 0.006, propensity AUC 0.51.
- Served CATE via a FastAPI `/predict_cate` endpoint and a Streamlit/Plotly dashboard; API tests and CI (ruff, pytest, image builds).

🔗 **Source:** github.com/PundarikakshNTripathi/Causal-DML

🔗 **HiveTorch: Federated Learning over gRPC and Kubernetes** | PyTorch, gRPC, protobuf, safetensors, Kubernetes, Prometheus, FedAvg

- Implemented FedAvg in-process and as a gRPC federation (protobuf, safetensors weights, server-side validation of session, round and tensor shapes); the networked run reproduces the in-process accuracy trace to 1e-6.
- Deployed as Docker Compose services and a Kubernetes Indexed Job (pod index picks the data partition) with Prometheus metrics and alerts; ran 1 server + 5 clients end to end on Minikube.
- Measured FedAvg on MNIST (20 clients, 3 seeds): 97.3% IID vs 96.4% at Dirichlet $\alpha=1$, 92.9% at $\alpha=0.1$ and 90.0% with 2-class shards.

🔗 **Source:** github.com/PundarikakshNTripathi/HiveTorch

🔗 **nanoDist: Data Parallelism and ZeRO-2 from Scratch** | Python, NumPy, TCP sockets, ring all-reduce, ZeRO-2, FP16, Hydra, Optuna, W&B

- Built ZeRO stage 2 on a from-scratch ring all-reduce: gradient reduce-scatter, Adam on each rank's 1/N parameter shard with 1/N optimizer state, parameter all-gather; matches single-process Adam (2 and 4 processes).
- Added FP16 compute on FP32 master weights with loss scaling, gradient accumulation and activation checkpointing; analytic accounting for a 1024→4096→1024 MLP: 105.97→33.84 MB per worker (–68%).

🔗 **Source:** github.com/PundarikakshNTripathi/nanoDist

Achievements

- **3rd on the public and private leaderboards (89,000+ teams), Amazon ML Challenge 2026** at the last check; macro-F0.5 0.991762. Led planning, coordination and review of a 3-person team's entity-resolution pipeline (1.73M businesses vs ≈10M noisy records): 13 lexical retrievers + fine-tuned e5 kNN (99.82% candidate recall), LightGBM A→B→C cascade (87 features, cross-encoder blend, 5-seed bag), calibration and expected-F0.5 set selection.
 - Diagnosed a validation/leaderboard gap (0.9896 vs 0.981) as unseen-country shift (France appears only in test; global IDF gave ≈42% of French tokens maximal weight); fixed with per-country IDF and pseudo-labels, adopting changes only if leave-one-country-out improved both ways (US→India 0.936→0.952).
- **2nd place, Masai × NoBroker HackArena 2025:** CivicAgent, an AI civic-complaint triage platform (Gemini Vision classification, APScheduler SLA tracking with email escalation, Supabase/PostGIS); proof of concept, no longer hosted.

Technical Skills

Languages: Python, C++20, CUDA C++, PTX (inline asm), Go, C, SQL, TypeScript, Bash

ML Frameworks: PyTorch, JAX, Transformers (Hugging Face), ONNX, Triton, scikit-learn, LightGBM, EconML, DoWhy

Systems and Infra: eBPF (LSM), AVX2 SIMD, CUDA (cp.async, CUB), DDP / ZeRO / ring all-reduce, gRPC, Docker, Kubernetes, CMake, GitHub Actions

Data and Backend: DuckDB, PostgreSQL, SQLite, FastAPI, Supabase, MLflow, Prometheus

Education

Babu Banarasi Das University

Lucknow, UP

B.Tech, Computer Science and Engineering (Artificial Intelligence)

Expected August 2027

- **Coursework:** Data Structures, Algorithm Design and Analysis, Database Management Systems, Linear Algebra, Probability, Statistics.
- **Self-study:** modern C++, CUDA, ML systems, causal inference.